Nihar Thampi

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Summary

Software and Data Engineer with a strong Python foundation, working across backend APIs, data pipelines, machine learning and LLM-powered applications. Experienced building REST/GraphQL services, ETL and analytics workflows, RAG and NLP systems and dashboards for financial, healthcare and security clients, deploying to AWS with a focus on reliability and fault-tolerant design.

Skills

Programming Languages:	Python, SQL (PostgreSQL, MySQL, SQLite)
Frameworks & APIs:	FastAPI, Flask, Flutter, REST / GraphQL APIs, JWT / OAuth2, RBAC
Data & Analytics:	Pandas, NumPy, Matplotlib, Power BI, ETL pipelines, data modelling, web scraping (Selenium, BeautifulSoup)
AI / ML:	PyTorch, TensorFlow, Hugging Face Transformers, OpenCV, EasyOCR; NLP, LLMs, RAG, LoRA / PEFT fine-tuning, prompt engineering
Tools & Cloud:	AWS, FAISS, ChromaDB, OpenAI / Gemini APIs, Git, Linux

Education

Vellore Institute of Technology (Chennai) Sep 2021 – Jul 2025
Bachelor of Technology in Computer Science and Engineering with Specialization in Artificial Intelligence and Machine Learning CGPA: 8.08

Experience

- **NNInfosec – Independent Consultant: Software Development** Jun 2026 – Present
  Built and shipped a full-stack Governance, Risk & Compliance (GRC) platform from a formal SRS plus a threat-intelligence automation service, working independently across backend, frontend and deployment.
  - Designed distributed, fault-tolerant REST and GraphQL APIs in Python (FastAPI, PostgreSQL) for the GRC platform, implementing JWT/MFA authentication, role-based access control, audit logging and database migrations.
  - Built CSV/Excel bulk-import pipelines with per-row validation for compliance data, and a React + TypeScript admin dashboard for compliance reporting and platform administration.
  - Automated customer-focused threat-intelligence IP blocking with a Python service that pulls attacker-IP data from CrowdStrike Falcon NGSIEM and VirusTotal APIs, scores each IP and maintains a continuously updated blocklist, running unattended as a Windows service.

- **Vault22 - Independent Consultant: Software Development** Mar 2026 – Jun 2026
  Designed and shipped production features across a multi-tenant wealth management platform, contributing to both the .NET 9 backend API and React dashboard within an agile team.
  - Developed RESTful backend services in C# (.NET 9), implementing MediatR-based CQRS handlers, domain business logic, EF Core migrations and PostgreSQL persistence across multiple financial modules.
  - Engineered multi-scheme authentication in C# (.NET 9) including OAuth2/JWT via Azure Entra ID and a one-time-token SSO flow, with tenant-scoped roles and permission enforcement and structured audit logging across partner integrations.
  - Built data-intensive React administrative dashboards featuring server-side filtering, paginated data grids, role-based access control and per-tab session isolation.

- **Fischer Jordan- Data Analyst** Mar 2025 – Apr 2026
  Joined as a Data Analyst Intern and converted to a full-time role after 4 months; collaborated with clients to define project requirements and delivered multiple data-driven solutions simultaneously across the financial and healthcare domains.
  - Built scalable Python backend services using FastAPI/Flask, Pandas, web scraping, scheduled ETL pipelines and PostgreSQL to automate ingestion, validation and transformation of heterogeneous datasets for production applications.
  - Developed a production-grade enterprise RAG chatbot using Python, PyTorch, FAISS and FastAPI, transforming multi-format unstructured documents into a searchable knowledge base with semantic retrieval and context-aware responses.

- Engineered low-latency trading and data automation systems using Python, AWS EC2/Lambda/EventBridge, PostgreSQL and AsyncIO, allowing concurrent processing, fault-tolerant retries and automated market-data workflows.
- Enhanced production prediction and scoring with Python decision models and LLM APIs for behavioral analysis and query routing, reducing therapist report analysis from ~1 hour to under 5 minutes.
- Engineered end-to-end data processing pipelines integrating Python web scraping (Selenium, BeautifulSoup), JSON/XML parsing, Gemini APIs and parallel processing with optimized LLM prompts, reducing data-entry turnaround from 4-5 days to 1-2 hours.

- **SnapitAI - Software Developer Intern** Oct 2024 – Dec 2024

  Deployed an AI-driven mobile application with real-time data pipelines and an LLM-based chatbot experience.
  - Built a Flutter mobile application integrating SERP and Llama LLM APIs to extract Google-services data and generate content, powering context-aware chatbots.
  - Deployed the application on Microsoft Azure virtual machines for scalability, reliability and performance.

Projects

- **AI-Powered Tax Compliance & Advisory Platform** *Python, Flask, React (Vite/TypeScript), ChromaDB, Hugging Face Transformers, Docker*

  A chatbot for South Africa's Income Tax Act that answers tax questions in plain language with a cited source for every point, and scans bank statements for likely deductions.
  - Built a Flask RAG service over a ChromaDB index of the Act with a local Hugging Face LLM, returning answers with per-point section, line and page citations through JWT-secured REST APIs consumed by a React (Vite/TypeScript) frontend with saved, searchable chats.
  - Developed a bank-transaction deduction scanner that batches CSV statements, retrieves supporting Act excerpts and returns JSON-validated classifications with cited legal basis and reasoning, with a local-LLM fallback.
  - Built an OCR ingestion pipeline (Tesseract + PyMuPDF) that digitises a ~1,000-page statute into overlapping chunks with batched embedding writes, plus a config-selectable LoRA fine-tuned model, containerised with Docker (CPU or CUDA 4-bit).

- **Invoice Management Web App (FairShare)** *Python, Flask, EasyOCR, SQLite, OpenAI API (GPT-4o-mini), React/TypeScript, Docker*

  An invoice-management app that extracts every field and line item from a photographed invoice or receipt, tracks what is unpaid or overdue, and splits shared bills item by item.
  - Built an OCR-to-LLM invoice pipeline: EasyOCR extraction with skew correction and line grouping, then GPT-4o-mini in JSON mode with a repair-and-retry loop, extracting vendor, dates, currency and line items behind Flask APIs, checked against the printed total.
  - Implemented item-level split logic (equal, percentage and custom) with idempotent recalculation and a rounding-reconciliation pass so per-person totals sum exactly to the invoice total.
  - Designed a SQLite relational model with schema migrations and JWT access/refresh-token auth, plus paid/overdue tracking, a per-currency dashboard and formula-safe CSV export in a React/TypeScript frontend.

- **Social Media Content Analyzer** *Python, FastAPI, React, PyTorch, Selenium, Transformers, FAISS, Docker*

  A tool that reads through thousands of YouTube comments and tells you what viewers think - sentiment, sarcasm, summaries and a chat window for follow-up questions.
  - Built a FastAPI + React/Vite analytics platform with dual-source ingestion - YouTube Data API v3 plus a headless Selenium/BeautifulSoup fallback - for resilient comment, transcript and metadata collection.
  - Built a background job orchestrator running transcript, sentiment, summarisation and indexing stages with status tracking, lazy model loading and device-aware (CUDA/MPS/CPU) inference.
  - Integrated SieBERT/BART-MNLI sentiment, DistilBART summaries and Mistral-7B sarcasm detection with a FAISS RAG chat, surfaced in a JWT-secured React dashboard with per-user data isolation.